REST API SPECIFICATION

Bug Panel API Reference & Workflow

Centralized Bug Tracking, Media Uploads, Resolution Verification & Audit Trails

Base: /api/bugs | Auth: Bearer JWT | Backend: Node/Express | Sept 2026

1. Architecture & Lifecycle Verification Workflow

The Bug Panel module provides enterprise bug reporting, triaging, media capture, collaboration threads, resolution verification, and immutable audit logs. All endpoints require Bearer JWT authentication via requireClearHire and requireAuth.

CRITICAL RULE - DEVELOPER CLOSURE RESTRICTION:
Developers CANNOT directly set a bug's status to CLOSED or CLOSED_VERIFIED. Developers submit a resolution via POST /:id/resolution (moving status to AWAITING_REPORTER_CONFIRMATION). Only the original reporter, an admin, or a manager can verify and close the bug.

Phase	Lifecycle Status	Primary Actor	Description & Rules
Step 1	`OPEN`	Any Authenticated User	Bug reported with title, description, attachments & source panel.
Step 2	`TRIAGED / IN_PROGRESS`	Manager / Lead / Dev	Severity & priority assigned; developer assigned to fix.
Step 3	`NEEDS_INFO`	Assigned Developer	Developer requests additional reproduction info from reporter.
Step 4	`AWAITING_CONFIRMATION`	Assigned Developer	Dev submits fix summary, commit URL, and verification proof.
Step 5a	`CLOSED_VERIFIED`	Original Reporter / Admin	Reporter verifies fix in staging/production and confirms closure.
Step 5b	`REOPENED`	Original Reporter	Reporter tests fix; if bug persists, rejection reopens bug to developer.
Override	`CLOSED_ADMIN_OVERRIDE`	Admin / Super-Admin	Administrative force-close bypassing reporter verification.

2. Bug Reports & Analytics Endpoints

POST /api/bugs/upload

Role: All Authenticated Users

Uploads up to 10 files (videos, screenshots, crash logs) with automated SHA-256 integrity checksum calculation.

REQUEST PAYLOAD:

```
- files: multipart/form-data array (max 10 files, combined max 250 MB)
- Images: JPEG, PNG, WEBP, HEIC (Max 20 MB each)
- Videos: MP4, MOV, WEBM, OGG, MKV (Max 150 MB each)
- Documents: PDF, DOCX, XLSX
```

RESPONSE (SUCCESS):

```
-> HTTP 201 Created: { items: [{ fileName, url, mimeType, size, duration, checksum, processingStatus }] }
```

GET /api/bugs/analytics

Role: All Authenticated Users

Calculates real-time KPIs, pending bug counts, acceptance rates, reopen rates, avg resolution hours, and top modules/devs.

RESPONSE (SUCCESS):

```
-> total: total bug count
-> pendingBugs, awaitingConfirmation, reopenedBugs, closedVerified
-> avgResolutionTimeHours: average hours from creation to resolution submission
-> acceptanceRate, reopenRate: percentages
-> topModules: [{ name, count }], topDevelopers: [{ name, count }], topReporters: [{ name, count }]
```

GET /api/bugs

Role: All Authenticated Users

Lists bug reports with multi-dimensional filtering, search, and pagination (default 25 items, max 100).

QUERY / PATH PARAMETERS:

```
- status: 'open' (default), 'closed', 'all', or specific status enum
- severity: 'low' | 'medium' | 'high' | 'critical' | 'all'
- priority: 'low' | 'medium' | 'high' | 'urgent' | 'all'
- company, module: filter by company name or software module
- developer, reporter: case-insensitive regex search on username
- q: multi-field text search (title, description, task, module, user)
- page: page number (default 1) | limit: items per page (default 25, max 100)
```

RESPONSE (SUCCESS):

```
-> HTTP 200 OK: { items: [BugSummary], pagination: { totalItems, totalPages, currentPage, limit } }
```

POST /api/bugs

Role: All Authenticated Users

Creates a new bug report, adds creator as participant, creates audit event, and dispatches notifications.

REQUEST PAYLOAD:

```
- title: string (required) | description: string (required)
- severity: 'low' | 'medium' | 'high' | 'critical' (default: 'medium')
- priority: 'low' | 'medium' | 'high' | 'urgent' (default: 'medium')
- company, module: string (optional organizational metadata)
- taskId: ObjectId (optional link to parent Task)
- assignedDeveloperId, assignedDeveloperName: string (optional initial assignment)
- attachments: array of upload objects (max 10)
- source: { panel: string, path: string } (reproduction origin)
```

RESPONSE (SUCCESS):

```
-> HTTP 201 Created: { item: BugReportObject }
```

GET /api/bugs/:id

Role: All Authenticated Users

Retrieves complete bug record including resolution and verification metadata. Automatically marks unread updates as read for the user.

QUERY / PATH PARAMETERS:

```
- id: MongoDB ObjectId of the bug report
```

RESPONSE (SUCCESS):

```
-> HTTP 200 OK: { item: FullBugReportObject }
```

PUT /api/bugs/:id

Role: Developer / Manager / Admin

Updates triage severity, priority, developer assignment, and status. Rejects unauthorized closure attempts.

REQUEST PAYLOAD:

```
- severity: 'low' | 'medium' | 'high' | 'critical'
- priority: 'low' | 'medium' | 'high' | 'urgent'
- assignedDeveloperId, assignedDeveloperName: string
- status: 'TRIAGED' | 'IN_PROGRESS' | 'NEEDS_INFO' (restricted from closed statuses for developers)
- title, description: editable by Admin and Manager roles only
```

RESPONSE (SUCCESS):

```
-> HTTP 200 OK: { item: UpdatedBugReportObject }
```

! NOTE: Attempting to set status to CLOSED_VERIFIED or CLOSED without admin/reporter role returns HTTP 403 Forbidden.

PUT /api/bugs/:id/close

Role: Admin / Manager / Original Reporter Only

Dedicated endpoint to verify and close a bug directly without a formal PR/release submission.

RESPONSE (SUCCESS):

```
-> HTTP 200 OK: { item: ClosedBugReportObject }
```

! NOTE: Developers are strictly rejected with HTTP 403 Forbidden.

3. Resolution & Verification Lifecycle Endpoints

POST `/api/bugs/:id/resolution` — **Role: Assigned Developer / Team Lead / Admin**

Developer submits fix proof and deployment details. Sets status to AWAITING_REPORTER_CONFIRMATION and notifies reporter.

REQUEST PAYLOAD:

```
- summary: string (required - explanation of root cause and fix)
- verificationPerformed: string (required - test cases & verification performed)
- releaseVersion: string (e.g., 'v1.4.2')
- deploymentEnvironment: string (e.g., 'Staging', 'Production')
- commitUrl, pullRequestUrl: string (links to version control)
- disposition: string (default: 'Fixed')
- attachments: array of fix proof screenshots/recordings
```

RESPONSE (SUCCESS):

```
-> HTTP 200 OK: { item: UpdatedBugReportObject }
```

POST `/api/bugs/:id/confirm-resolution` — **Role: Original Reporter / Admin Only**

Reporter verification gate. Confirms fix or rejects it to reopen the bug.

REQUEST PAYLOAD:

```
- confirmed: boolean (true = accept fix, false = reject fix)
- feedback: string (optional feedback when confirming)
- rejectionReason: string (mandatory if confirmed is false)
```

RESPONSE (SUCCESS):

```
-> HTTP 200 OK: { item: UpdatedBugReportObject }
```

! NOTE: If confirmed == true: Status -> CLOSED_VERIFIED, logs REPORTER_CONFIRMED event.
! NOTE: If confirmed == false: Status -> REOPENED, logs REPORTER_REJECTED event and notifies dev.

POST `/api/bugs/:id/request-info` — **Role: Assigned Developer / Admin**

Developer requests clarification or reproduction steps from reporter. Moves status to NEEDS_INFO and notifies reporter.

REQUEST PAYLOAD:

```
- note: string (details on information needed from reporter)
```

RESPONSE (SUCCESS):

```
-> HTTP 200 OK: { item: UpdatedBugReportObject }
```

POST `/api/bugs/:id/admin-override` — **Role: Admin / Super-Admin Only**

Forces bug closure with administrative override, bypassing reporter confirmation.

REQUEST PAYLOAD:

```
- reason: string (justification for administrative force closure)
```

RESPONSE (SUCCESS):

```
-> HTTP 200 OK: { item: UpdatedBugReportObject }
```

! NOTE: Sets status to CLOSED_ADMIN_OVERRIDE and logs immutable ADMIN_OVERRIDE event.

4. Collaboration Threads & Audit Log Endpoints

GET `/api/bugs/:id/comments` — **Role: All Authenticated Users**

Fetches all threaded comments and discussion messages for a bug sorted chronologically (oldest first).

RESPONSE (SUCCESS):

```
-> HTTP 200 OK: { items: [BugCommentObject] }
```

POST `/api/bugs/:id/comments` — **Role: All Authenticated Users**

Adds a comment to the bug discussion thread. Parses @mentions and sends alerts to mentioned users and assignees.

REQUEST PAYLOAD:

```
- content: string (max 10,000 characters)
- attachments: array of uploaded file objects (max 10)
- userAvatarUrl: string (optional user avatar URL)
```

RESPONSE (SUCCESS):

```
-> HTTP 201 Created: { item: BugCommentObject }
```

PUT `/api/bugs/:id/comments/:commentId` — **Role: Comment Author Only**

Edits comment content. Strictly enforced within a 5-minute window from posting.

REQUEST PAYLOAD:

```
- content: string (required updated text)
```

RESPONSE (SUCCESS):

```
-> HTTP 200 OK: { item: UpdatedCommentObject }
```

! NOTE: Editing after 5 minutes returns HTTP 400 Bad Request.

DELETE `/api/bugs/:id/comments/:commentId` — **Role: Comment Author / Admin**

Soft-deletes a comment. Replaces content with 'Comment Removed' and purges attachments to preserve audit integrity.

RESPONSE (SUCCESS):

```
-> HTTP 200 OK: { item: DeletedCommentObject }
```

GET `/api/bugs/:id/events` — **Role: All Authenticated Users**

Retrieves complete immutable activity logs for compliance, QA tracking, and workflow timeline visualization.

RESPONSE (SUCCESS):

```
-> HTTP 200 OK: { items: [BugEvent] }
-> Event Types: BUG_CREATED, ASSIGNED, REASSIGNED, PRIORITY_CHANGED, SEVERITY_CHANGED,
-> STATUS_CHANGED, COMMENT_ADDED, COMMENT_EDITED, COMMENT_DELETED, RESOLUTION_SUBMITTED,
-> REPORTER_CONFIRMED, REPORTER_REJECTED, BUG_REOPENED, ADMIN_OVERRIDE, INFO_REQUESTED
```

5. Role-Based Access Control (RBAC) Matrix

Operation / Endpoint	Reporter	Developer	Manager	Admin
Submit Bug (POST /api/bugs)	Full Access	Full Access	Full Access	Full Access
Upload Files (POST /upload)	Max 10 files	Max 10 files	Max 10 files	Max 10 files
Assign Developer / Triage	Forbidden	Self-Assign Only	Full Access	Full Access
Submit Resolution	Forbidden	Full Access	Full Access	Full Access
Verify / Reject Fix	Original Reporter	Forbidden	Forbidden	Override Access
Direct Close (PUT /:id/close)	Original Reporter	Forbidden (403)	Full Access	Full Access
Admin Force Override	Forbidden	Forbidden	Forbidden	Full Access
Post Comments / Mentions	Full Access	Full Access	Full Access	Full Access
Edit Comment (5 min window)	Own Comment	Own Comment	Own Comment	Own Comment
Delete Comment	Own Comment	Own Comment	Own Comment	Any Comment
View Audit Events	Full Access	Full Access	Full Access	Full Access